Final results digest — forecast-informed vision-pool provisioning

Simulation results (as requested) + physical validation (new)

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2026-07-28 — thesis build: 135 pp, 0 errors; results traceable to pinned artifacts

The chain being graded

Encoder-specific demand signal → provisioning-horizon probabilistic forecast → replica schedule under startup delay → window-level SLA/resource grading against matched controls (static optimum, reactive), on the public Azure multimodal trace (one million requests, seven days), under a single pre-frozen replay protocol.

1. Final simulation results

Premise (the signal gap is real). Request rate misrepresents vision demand: constant-fraction explains R^2=0.12 of the image rate vs. 0.65 of decode demand; image arrivals are an order of magnitude burstier; the two signals' extreme minutes are essentially disjoint. In simulation the pools' saturation knees sit an order of magnitude apart, and inside the burst window the encoder pool misses its tail bound *at the trace's own recorded rate* — invisible to rate/utilization monitors.

Forecastability. Zero-shot time-series foundation models beat naive/EWMA/Holt-Winters in **20/20** cells; the vision signal is the most persistent (autocorrelation half-life 69 min vs. 9 for requests); the p90 band edge is empirically calibrated (coverage 0.88–0.90), holding across the 12-step horizon and inside burst windows.

Closed loop — the headline. Over the 5.60-day, 749,601-request evaluation span, the p90-forecast schedule holds the encoder tail bound at **0.62**× the deployable static fleet's provisioned worker-time (**669.7 vs. 1,074.8** worker-hours). Forecast and reactive delay quantiles coincide through p99 over the span; the latency advantage lives in the one-minute onsets. In a reactive-favorable post-hoc floor sweep (forecast fixed; no symmetric forecast-plus-floor sweep), common-warm M=2 costs **1.20**× the forecast's worker-time; shipped strict M=6 gives a non-symmetric **1.67**× sensitivity. A perfect-information reference at 0.292× marks remaining signal-quality headroom.

Language pool (secondary). The forecast-driven language policy holds its SLA only through tail-aware capacity conversion; at its published operating point it is near break-even vs. static (0.965× time-matched, 1.03× request-matched). Week-scale language grading: the reactive span TTFT p99 breaches (1196.6 ms vs. forecast 651.1 ms, consistent across three forecast models) with 37.4% less total TTFT mass for forecast.

Live-stack check. Both burst-window verdicts reproduce in both directions through Dynamo's *real* control plane (mocked engines, single run per policy): forecast passes (695 vs. 651 ms offline), reactive fails (3,531 vs. 3,495 ms).

2. Physical validation (the plus — new since we spoke)

- **Cost model on silicon:** encoder costs collected on RTX 5090 (locked clocks; dataset submitted upstream to NVIDIA aiconfigurator as a draft PR) and validated against the serving engine's *own* vision tower: within **12.8%** across the entire $N \leq 64$ batching regime (attention operator within ≈3%). Re-running the loop with measured costs moves the headline p99 only 589→606 ms.
- **Controller dispatches replayed on silicon** (five predeclared reps/arm, exact frozen dispatch, membership-exact): on the fastest-onset burst the simulated verdict survives with a ≈29× margin — forecast 782.455 ± 0.270 ms (inside the 1 s bound) vs. floorless reactive $22{,}565.3 \pm 0.9$ ms. Every repetition reconstructs the banked simulation percentiles within the declared numerical tolerances before retiming.
- **Opposite regime too:** on a calm 24-hour slice the simulation's structure is preserved on silicon — static lowest, forecast/reactive near-tied, and every arm passes under the serving engine's measured service times. (A slower HuggingFace-eager stack inflates absolutes 1.5–1.6×; reported as an engine sensitivity case, not a bound.)
- **Integrity:** the negative-gate suite rejects all 18 of its injected corruptions loudly, from a clean checkout; raw evidence is pinned at immutable commits with model hashes, the applicable container digest, and the explicit no-container case.

Verification state

Results are traceable to pinned artifacts; the 0.623 headline and comparator cells recompute from a detached clean checkout; role-scoped semantic gates bind the load-bearing claims with adversarial self-tests, and the 65-check delivery suite is green.